

# ■ Machine Learning Formula Handbook (Running PDF)

## ■■ Date: 08/12/2025 – Linear Regression & Gradient Descent

Regression equation, MSE, gradient descent updates.

## ■■ Date: 09/12/2025 – Tree, SVM, Logistic, Scaling

Entropy, Gini, Sigmoid, MinMax scaling.

## ■■ Date: 10/12/2025 – Naive Bayes

Bayes theorem, Gaussian distribution.

## ■■ Date: 13/12/2025 – Kernel Functions

Linear, Polynomial, RBF, Sigmoid kernels.

## ■■ Date: 13/12/2025 – Regularization & Lasso

L1 and L2 regularization objective functions.

## ■■ Date: 13/12/2025 – KNN

Distance metrics.

## ■■ Date: 13/12/2025 – Classification Metrics

Precision, Recall, F1-score formulas.

## ■■ Date: 13/12/2025 – PCA

Covariance matrix, eigenvalues, eigenvectors.

## ■■ Date: 13/12/2025 – Bias & Variance

Trade-off summary.

## ■■ Date: 13/12/2025 – Bagging

Bootstrap aggregation logic.

## ■■ Date: 13/12/2025 – Image Preprocessing (OpenCV)

Normalization  $x' = x/255$ , resizing concept.